1..LTI SYSTEM

```
clc;
clear;
close all;

%% Given sequences
h1 = [1/2 1/4 1/2];

% h2[n] = h3[n] = delta[n] + 0.5delta[n-1] + delta[n-2]
h2 = [1 0.5 1];
h3 = [1 0.5 1];

% h4[n]
h4 = [1 1 1];

%% Overall impulse response
% Upper path = h2
% Lower path = h3 * h4
% Both paths are added, then convolved with h1

h_lower = conv(h3,h4);
h_parallel = add_sequence(h2,h_lower);
h = conv(h1,h_parallel);

disp('Overall impulse response h[n]:');
disp(h);

%% Plot
figure;
stem(0:length(h)-1,h,'filled');
grid on;
xlabel('n');
ylabel('h[n]');
title('Overall Impulse Response h[n]');

%% Local function for adding unequal length sequences
function y = add_sequence(x1,x2)

    L = max(length(x1),length(x2));

    x1 = [x1 zeros(1,L-length(x1))];
    x2 = [x2 zeros(1,L-length(x2))];
```

```
    y = x1 + x2;
end
```

(ii) Without built-in conv — using for loop

```
   clc;
clear;
close all;

h1 = [1/2 1/4 1/2];
h2 = [1 0.5 1];
h3 = [1 0.5 1];
h4 = [1 1 1];

%% Manual convolution function
h_lower = myconv(h3,h4);

L = max(length(h2),length(h_lower));

h2_pad = [h2 zeros(1,L-length(h2))];
h_lower_pad = [h_lower zeros(1,L-length(h_lower))];

h_parallel = h2_pad + h_lower_pad;

h = myconv(h1,h_parallel);

disp('Overall impulse response h[n]:');
disp(h);

figure;
stem(0:length(h)-1,h,'filled');
grid on;
xlabel('n');
ylabel('h[n]');
title('Overall Impulse Response');

%% Manual convolution
function y = myconv(x,h)

    y = zeros(1,length(x)+length(h)-1);

    for i = 1:length(x)
        for j = 1:length(h)
```

```
            y(i+j-1) = y(i+j-1) + x(i)*h(j);
        end
    end

end
```

2…Autocorrelation of deterministic sequences

```
       clc;
clear;
close all;

%% (i) Rectangular pulse
x1 = ones(1,10);

[r1,lags1] = xcorr(x1);

figure;
stem(lags1,r1,'filled');
grid on;
xlabel('Lag');
ylabel('Autocorrelation');
title('Autocorrelation of Rectangular Pulse');


%% (ii) Triangular pulse
x2 = [1 2 3 4 5 4 3 2 1];

[r2,lags2] = xcorr(x2);

figure;
stem(lags2,r2,'filled');
grid on;
xlabel('Lag');
ylabel('Autocorrelation');
title('Autocorrelation of Triangular Pulse');


%% (iii) Sinusoidal signal
N = 50;
n = 0:N-1;

x3 = sin(2*pi*0.1*n);
```

```
[r3,lags3] = xcorr(x3);

figure;
stem(lags3,r3,'filled');
grid on;
xlabel('Lag');
ylabel('Autocorrelation');
title('Autocorrelation of Sinusoidal Signal');


%% (iv) Exponential signal
x4 = (0.8).^n;

[r4,lags4] = xcorr(x4);

figure;
stem(lags4,r4,'filled');
grid on;
xlabel('Lag');
ylabel('Autocorrelation');
title('Autocorrelation of Exponential Signal');
```

3…Autocorrelation and cross-correlation of random sequences

```
        clc;
clear;
close all;

N = 1000;

%% Generate random sequences
x = randn(1,N);
y = randn(1,N);

%% Autocorrelation of x
[rx,lagsx] = xcorr(x);

figure;
plot(lagsx,rx);
grid on;
xlabel('Lag');
ylabel('Rxx');
title('Autocorrelation of Random Sequence x[n]');
```

```
%% Autocorrelation of y
[ry,lagy] = xcorr(y);

figure;
plot(lagy,ry);
grid on;
xlabel('Lag');
ylabel('Ryy');
title('Autocorrelation of Random Sequence y[n]');


%% Cross-correlation
[rxy,lagsxy] = xcorr(x,y);

figure;
plot(lagsxy,rxy);
grid on;
xlabel('Lag');
ylabel('Rxy');
title('Cross-Correlation of Random Sequences');
```

4…Radar time-delay estimation

```
    clc;
clear;
close all;

%% Parameters
N = 200;
a = 0.9;
D = 20;

% 13-point Barker sequence
barker = [1 1 1 1 1 -1 -1 1 1 -1 1 -1 1];

%% Generate x[n] of length 200
x = repmat(barker,1,ceil(N/length(barker)));
x = x(1:N);

%% Noise variance
sigma2 = 0.01;

%% Generate Gaussian noise
```

```
v = sqrt(sigma2)*randn(1,N);

%% Generate delayed received signal
y = zeros(1,N);

for n = D+1:N
    y(n) = a*x(n-D) + v(n);
end

%% Plot x[n] and y[n]
figure;

subplot(2,1,1);
stem(0:N-1,x,'filled');
grid on;
xlabel('n');
ylabel('x[n]');
title('Transmitted Signal x[n]');

subplot(2,1,2);
stem(0:N-1,y,'filled');
grid on;
xlabel('n');
ylabel('y[n]');
title('Received Signal y[n]');


%% Cross-correlation
[rxy,lags] = xcorr(y,x);

% Consider positive lags up to 59
index = (lags >= 0 & lags <= 59);

figure;
plot(lags(index),abs(rxy(index)),'LineWidth',1.5);
grid on;
xlabel('Lag l');
ylabel('|r_{xy}(l)|');
title('Cross-Correlation for Delay Estimation');

%% Estimate delay
[~,idx] = max(abs(rxy(index)));
lag_values = lags(index);
```

```
estimated_delay = lag_values(idx);

disp(['Actual delay D = ',num2str(D)]);
disp(['Estimated delay = ',num2str(estimated_delay)]);
```